

**CPE 381 CLASSWORK 05: CUBIC SPLINE**

Student Name (as in Canvas): \_\_\_\_\_

A Number: \_\_\_\_\_

Points: 10    Duration: 10 minutes    Due: September 09, 2026 (In-Class)

Consider signal samples as  $(t_1, x_1), (t_2, x_2), \dots, (t_{n+1}, x_{n+1})$ .

1. For  $n + 1$  samples in a signal, if we were to interpolate using cubic spline, how many spline interpolant we would need? **(3 Points)**

**Answer** $n$  spline interpolant.

2. What's free boundary conditions in cubic spline formulation? **(3 Points)**

**Answer**

Assuming  $P_1$  as the first interpolant and  $P_n$  as the last interpolant in a cubic spline interpolation fitting on signal values, the free boundary condition states that

- $P_1''(t_1) = 0$  and  $P_n''(t_{n+1}) = 0$

where  $t_1$  is the first time stamp, and  $t_{n+1}$  is the last timestamp.

3. What's the advantage of cubic spline interpolation over linear interpolation? **(4 Points)**

**Answer**

- (a) Cubic spline provides twice differentiable smooth fitting of a signal on the signal values for interpolation.

However, one should note that cubic interpolation is global and adding or removing one data point changes the interpolant coefficients, while linear interpolants are purely local and are unaffected by adding or remove data points.